Smart Fisher Lanka is an AI-based decision support system designed for Sri Lankan fishermen

Project ID: 25-26J-177

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

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Research Component: AI-Driven Fish Quality Grading and Weight Estimation Module

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor

(Prof. Pradeep Abeygunawardhena)

28/08/2025

Date

Signature of the co-supervisor

(Mr. Ashvinda Iddamalgoda)

2025/08/28

Date

Abstract

The Sri Lankan fishery industry is gravely threatened in maintaining consistent fish quality levels, with an economic implication and loss of market competitiveness. Hand grading of conventional fish is subjective, time-consuming, and plagued by human error, and its effect primarily falls on the small-scale fishermen and exporters who have no means of access to sophisticated quality assessment equipment.

This project envisions the deployment of a lightweight, real-time AI-driven fish quality grading system based on MobileNet CNN architecture integrated within a mobile application. The system will classify fish freshness, damage extents, and general quality grades from image analysis of mobile phone camera-captured photos immediately after harvest. Utilizing transfer learning techniques with MobileNetV2 and deploying the trained model to TensorFlow Lite, the solution will possess offline functions critical to maritime environments with limited connectivity.

The system suggested completes the principal void of available, impartial fish quality estimation tools designed for Sri Lankan species. Unlike existing global solutions, the system will be trained on local species datasets and customized for fishermen's and exporters' applications with varying technical proficiency. The solution will also contain empirical volume modeling to derive fish weight from density measurement, providing additional value for business use.

Expected outcomes are improved standardization of fish quality, minimized post-harvest losses, improved market prices for quality fish, and increased competitiveness of Sri Lankan fish exports in foreign markets. The mobile-based character of the system ensures access and real-time decision-making ability for stakeholders in the fishing value chain.

Keywords: Deep Learning, Fish Quality Grading, Weight Estimation, Length-Weight Relationship, Image Processing, Sri Lankan Fisheries, Real-Time Inference, Computer Vision, Mobile AI

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1. INTRODUCTION

1.1 Background and Literature survey

Sri Lanka's fisheries industry remains a significant pillar of national food security, rural livelihood, and economic resilience. The sector constitutes vast marine, freshwater, and aquaculture resources that supply the nutritional needs of millions through important protein and micro-nutrients that form the core of Sri Lankan cuisine. It provides a wide socio-economic base, engaging various communities and value chain stakeholders spread across the coastal belt and inland waters. The economic and ecological place of the fisheries sector is profound, uniting tradition with emerging technological adoption in a search for future development. Government and associated agencies always stress judicious management of fisheries to optimize productivity as well as sustainability, respecting the sector's capacity for faster development and export competitiveness under rationalized, data-driven platforms.[1]

Despite that, consistent technology and quality control problems limit industry potential. One of the significant issues is post-harvest loss, which has been estimated to affect up to 40% of the catch in some multi-day fisheries due to deterioration in storage, transport, and sales. Most fish are inspected visually by sensory inspection—graders using vision by observing such aspects as gill color, eye clearness, skin texture, and odor. This ancient system, though subjective, is heavily influenced by environmental condition, grader fatigue, or incentives in the market and thus leads to variable grading, price disputes, and wasteful rejection of potentially edible fish.[2],[3]

In much the same way, similar issues are plaguing global fisheries, and as a reaction, contemporary computer vision and artificial intelligence (AI) are revolutionizing quality control. Convolutional Neural Networks (CNNs), particularly light-weight models such as MobileNet, have been shown to perform precise image-based classification, even outperforming human graders in terms of standardization and efficiency.[4] MobileNet's characteristic depth wise separable convolutions make efficient operation on resource-limited platforms possible essential for mobile deployment in the remote fishery context. Transfer learning even accelerates performance gains in specialized small sets by adapting pre-trained models to large public image collections like ImageNet to perform rapid tailoring for local species.[5]

The literature reports that most of the advanced AI quality assessment systems are built on the datasets of temperate species (like salmon or cod) under laboratory-level lighting and imaging conditions.[6],[7],[8] This leaves us with an applicability gap: AI models built in foreign idealized environments rarely carry over into the variety of tropical species, catch conditions, and real trading conditions in Sri Lanka. Attempts at using live, mobile AI grading applications remain largely in prototype form, too often requiring expensive sensors, cloud connectivity, or controlled environments.

Therefore, most small-scale fishermen who constitute the backbone of the industry have no practical, affordable, and durable means for objective fish quality evaluation, with the result being inefficiency and inequality in income.[9]

Latest annual reports highlight the need for upgradation of technology, transparency in markets, and improved post-harvest management to support long-term sustainability as well as export growth. Government and research organizations such as the Department of Fisheries and Aquatic Resources (DFAR) and NARA look out for solutions that are context-aware, strong and user-friendly for everyone. In short, everything needed for digital transformation is ready, but no field-tested, species-specific, multi-parameter AI grading solution exists proven under real Sri Lankan operating conditions.[1]

1.2 Research Gap

Though fisheries are an economic mainstay in Sri Lanka employing over half a million workers and enabling national food security the sector is faced with chronic and interconnected technology, operational, and environmental issues that directly hinder modernization and sustainable development.[10]

To begin with, the problem of post-harvest quality loss is acute: Large proportions of Sri Lanka's marine fish catch (40% or more for multi-day fisheries) are believed to suffer quality loss in handling, storage and sale.[11] Losses are due to inadequate storage, delayed unloading, and a lack of rapid, objective grading at the point of first sale. Losses not only reduce productivity and profit but undermine both export potential and local nutrition.[12]

Available digital and AI-based solutions are not context-fit: Most foreign research, applications, and lab-level algorithms for fish quality utilize western, temperate species datasets (salmon, cod).[13] Such models are not generalizable to Sri Lanka's tropical species (Thora, Kelawalla, Spanish mackerel, and reef fish). There are morphological differences and differences in spoilage, and hence the imported AI models misclassify or mis-grade the local fish.

Technology solutions remain inaccessible and impractical for most stakeholders: State-of-the-art fish grading systems in literature require expensive hardware, cloud connectivity, controlled lighting, and specialist maintenance. Such systems are just not affordable for small-scale fishers and local market vendors of Sri Lanka who have limited capital and infrastructure.

No current system addresses multi-label grading, real-world deployment, and offline operation all at once:

Existing approaches are single attribute (species or freshness), and do so in isolated modules, not generating an actionable quality profile where needed. Most of the research assumes stable internet, cloud APIs, and controlled environments assumptions collapsing at sea, on beaches, or in open markets. More critically, no peer-reviewed work has validated mobile AI grading for Sri Lankan species in real local environments e.g., noisy ports, low lighting, mixed catch.[3]

Ecological, regulatory, and climatic stresses compound the disparity: The sector is faced with climate-driven species migration, infrastructure vulnerability, and illegal, unreported, and unregulated (IUU) fishing risks all calling for rapid, data-backed response and traceable quality documentation. Without the right on-device, multi-parameter grading functionality, the sector cannot ensure transparency, safety, or competitiveness.[14],[15]

1.3 Research Problem

The Sri Lankan fisheries industry greatly needs an integrated, objective, and cost-effective technology for grading fish quality, exactly fitted to the multi-faceted situation of tropical small-scale fisheries.

Reliance on subjective, manual appraisal leads to inconsistent quality at sale, repeated price lacks of agreement and ingrained information asymmetry that hurts small-scale fishers the most. This perpetuates a system of missed incomes, livelihoods laid bare, and low market transparency.

Existing automated tools are out of reach: They require significant initial investment, infrastructure, or connectivity unavailable outside of big cities or landing sites. Their failure to adapt to local species, unpredictable environmental conditions, and existing workflow guarantees that most AI grading tools remain unused or unusable by legitimate stakeholders the very ones most exposed to economic, climatic, and regulatory stress.

The research challenge is thus as much technical and practical as social designing durable multi-label grading of tropical fish with mobile AI as it is solving how to get to a system that is embedded in fisher workflows from catch to sale, applies across a variety of species and ecosystems, and is offline.

The solution will have to deliver not only laboratory accuracy, but working empowerment of Sri Lankan fishermen and market actors, supporting evidence-based decision-making, waste minimization, ensuring fair market prices, and facilitating sustainable growth of industry.

2. OBJECTIVES

2.1 Main Objectives

The primary objective of this project is to design, develop, and deploy a light, real-time, mobile-enabled fish quality grading system specifically for the Sri Lankan fisheries sector. Leveraging advanced deep learning algorithms, namely the MobileNetV2 model, our system will aim to benefit diverse stakeholders ranging from small scale local fishermen to exporters and market vendors. Using nothing more than the camera of a smartphone, this app will provide objective, instant, and accurate assessments of fish quality, freshness, physical flaws, and rough size that typically involve expert human inspection or costly lab analysis.

With this AI-powered innovation, the project solves several major industry problems such as reducing post-harvest loss through inconsistent quality grading, optimizing price transparency by removing subjective judgment from transactions, and optimizing traceability and efficiency for supply chains. It should render advanced quality control available to everyone and democratize even complex grading for even sparsely located or resource-constrained users who lack access to expensive equipment or technical expertise.

The system would be optimized and made for offline use, allowing real-time inference and grading within the remote coastal regions or on the ocean, where minimal or no internet access exists. This makes it possible for the technology to merge efficiently with existing fisherman workflows without adding to operational complexity or dependency on outside infrastructure.

By providing reproducible and standardized quality grades with data-driven confidence, the project aims to promote economic empowerment of fishermen, enabling them to negotiate better prices and plan catches better. It also helps with sustainability goals through minimizing waste, enabling smart fishing and post-catch decision-making that is targeted towards market demands as well as regulatory compliance.

Lastly, this effort is not merely technological innovation but also societal transformation through improved livelihoods, diminished economic risk, and modernization of one of Sri Lanka's most vital economic sectors through actionable, data-driven solutions.

2.2. Specific Objectives

- To develop an effective MobileNetV2-enacted computer vision model, trained and tuned to Sri Lankan commercial species, which can grade fish freshness, physical damage and type from post-harvest images.
- To make available a user-friendly mobile app (Flutter/TensorFlow Lite) that facilitates instantaneous, offline grading and assessment of individual catches of fish even in rural or offshore locations.
- To include a strong way of fish weight estimation by image-derived measurements to enable objective determination of yield and economic value.
- To validate the AI grading tool against expert graders and existing grading systems, with results to survive in actual Sri Lankan marketplaces and landing sites.
- To ensure that the solution is scalable, cost-effective, and accessible to both small-scale fishers and larger industry players, paving the way for broader digital transformation across the industry.

2.2.1. User Requirements

- The app must function flawlessly on low-cost, readily available Android phones.
- Fishers, suppliers, and retailers must have the ability to photograph an item and get instant real-time detailed grading (species, freshness, visible defects, estimated weight) with no technical expertise or additional hardware.
- The app must function in coastal, rural, or market environments, with multiple language support (Sinhala, Tamil, English) and uncomplicated step-by-step instructions.
- The system needs to generate confidence scores, explicit explanations of grading outcomes, and facilitate record keeping or convenient reporting to meet regulatory or export requirements.

2.2.2. Functional Requirements

- Enable users to take or upload fish pictures and receive automatic, multi-label grading (freshness, damage).
- Perform real-time image processing and MobileNetV2-based analysis only offline, i.e., without cloud services.

- Enable users to input reference measurements (for weight estimation) or use onboard image calibration feature if available
- Save grading results locally and provide export options (CSV, PDF, or cloud sync in the case of online) for traceability, audit, or sale documentation.
- Provide a natural UI/UX with minimal in the form of technical barriers, e.g., visual feedback and usage tips.
- Facilitate feedback collection for ongoing model refinement and end-user support.

2.2.3. Non-Functional Requirements

- Performance: The evaluation and grading process should return results within less than 3 seconds per image, even for less-capable hardware.
- Robustness: The application must work correctly under diverse real-world conditions, e.g., varying illumination, messy background, and differing fish presentation angles.
- Usability: Interfaces and workflows should minimize user input, using defaults, pre-populated form fields, and error-proofed instructions.
- Offline Capability: Inference of the whole model, grading, and storage of outcomes should be facilitated without access to live internet.
- Security and Privacy: User-provided images and information should be stored securely, accessible only to the user unless explicit sharing or export is chosen
- Accessibility: Touch-friendly UI support, local language support, and simple onboarding are essential to inclusivity.
- Extensibility: The system must be designed to allow easy upgrades (new species, grading rules, or features) and integration with future traceability or regulatory services.

3. METHODOLOGY

The project methodology is structured to create systematically, train, deploy, and validate a mobile AI-powered fish quality grading system for optimal performance in Sri Lankan fisheries.

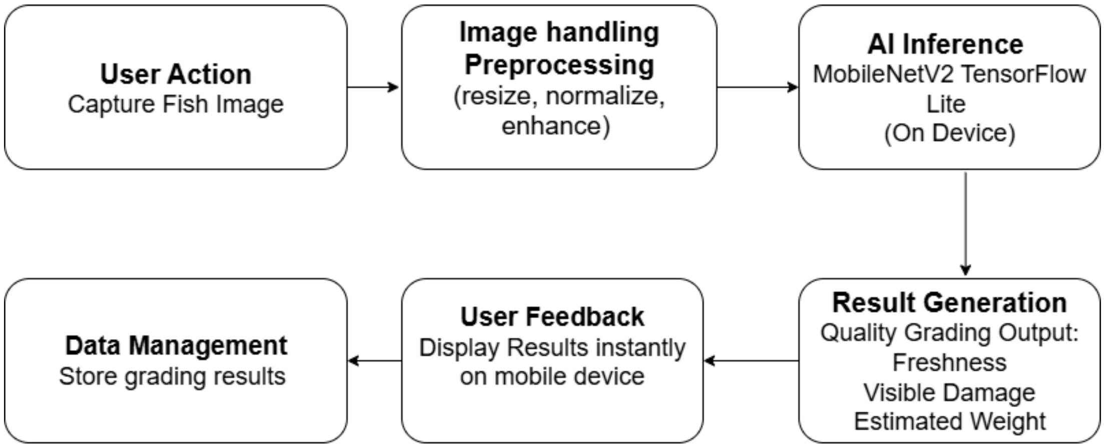

Figure 3.1: System flow diagram

3.1 Key Stages of the Process

3.1.1 Data annotation and collection

- Data Collection: Sample images representing economically important Sri Lankan fish species such as Yellowfin Tuna (Thora) and Yellowstripe Trevally (Kelawalla) will be sampled from local fish markets, landing sites, and fishing vessels.
- Annotation: Experienced fish graders will annotate each image with corresponding labels such as species, freshness grade, types of physical damage, and size references to create a robust, multi-label database.
- Environmental Variability: The dataset will have images captured under varying illumination, background, and pose to make the model robust and generalizable under varying field conditions.

3.1.2 Preprocessing and augmentation

- Images will be preprocessed to render them of uniform size, luminance, and contrast, and to segment fish features from the image background.
- Augmentation techniques like rotations, flips, color jittering, and occlusion simulation will synthetically vary the training set for improving model robustness against varying conditions in actual deployment.

3.1.3 Building and training the model

- Base Model: MobileNetV2, a lightweight convolutional neural network deployable on mobile platforms, will be employed as the base.
- Transfer Learning: Employing pre-trained weights from the ImageNet dataset will accelerate training on the specific fish dataset.
- Multi-Label Classification: The model will be trained to perform several tasks simultaneously damage detection, and freshness grading.
- Weight Estimation Regression: A second regression will estimate fish weight based on image data normalized against morphometric data specific to the species.

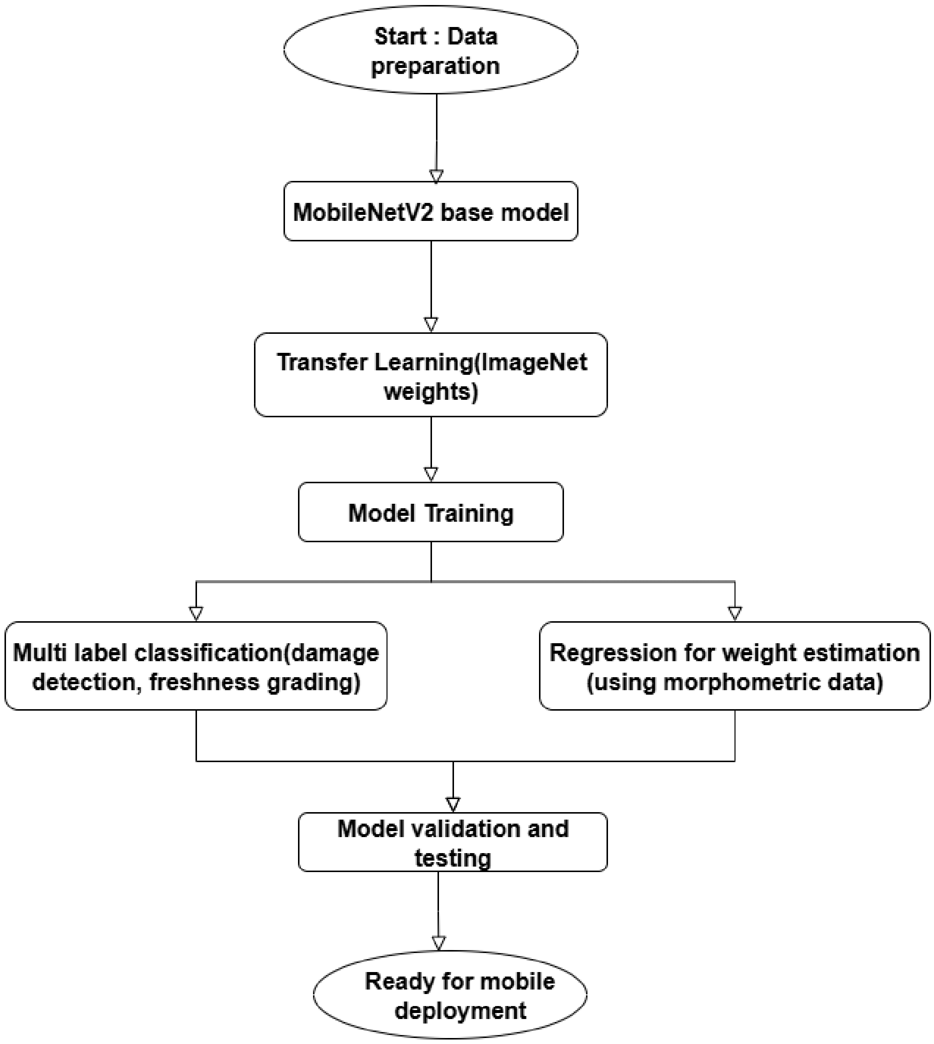

Figure 3.2: Model training diagram

3.1.4 Weight estimation methodology

- Users will take pictures of fish with a calibrated known size object (e.g., ruler, reference marker) in close proximity to the fish to create scale and enable accurate conversion of measurement.
- Sophisticated image processing techniques are employed by the system to segment the fish from the background and recognize anatomical landmarks (i.e., the snout and tail fork) to accurately gauge the length of fish.

- Pixel width and length measurements are converted to real-world dimensions through the use of the calibration reference for consistent measurement regardless of position or distance from the camera.
- Weight of fish is initially estimated from the generally accepted empirical length-weight relation for the species provided by:

$$\mathbf{W = a * L^b}$$

K is a species-specific constant usually ranging from 700 to 900.

- For greater precision, the system estimates girth—the circumference around the thickest part of the fish's body—by estimating width in images and calculating girth as approximately twice the width. This allows for the use of the length-girth empirical formula:

$$\mathbf{Weight = (Length * Girth^2)/2}$$

where W is weight, L is length, and a, b are species-specific constants derived from regionally calibrated biological data.

- To get more precise estimates, the model incorporates morphometric information by approximating body width at a set of standardized points along the fish's body to estimate body volume. Volume is calculated by summing elliptical cross-sectional area multiplied by segment lengths.
- Estimated volume can be made even more accurate by multiplying estimated volume by mean species-specific tissue density to capture variation in body condition that might not be detected by length alone.
- The system integrates machine learning regression models and paired image-measurement-ground truth training data to enhance and optimize traditional formula outcomes, adapting dynamically to species morphology and environmental factors.
- The multi-parameter approach is optimized for computational efficiency for mobile deployment as well as scientific precision, enabling feasible use in Sri Lankan fisheries with research-grade reliability.

3.1.5 Model deployment and optimization

- The model will be compressed and optimized (quantization, pruning) for on-device, real-time inference.
- It will be built to TensorFlow Lite and distributed in a Flutter mobile application executable on Android and iOS operating systems.
- Offline capability will be prioritized to enable grading feature in connectivity-low situations.

3.1.6 Mobile application development

- The user interface shall guide the user through image capture, grading outcome interpretation, and export or cloud syncing optionally.
- Multi-language interface support (Sinhala, Tamil, English) and user-friendliness will make the application accessible to fishers, traders, and exporters of all skill levels.

3.1.7 Validation and field testing

- Model output will be compared with expert human grader judgment at landing sites and market sites.
- User experience, environmental robustness, and performance metrics will drive ongoing improvement.
- Validation trials will incorporate a range of fish species and realistic usage conditions to validate the system's operational effectiveness.

3.2 Tools and Technology

The project employs deep learning platforms such as TensorFlow and Keras to make AI models, with TensorFlow Lite enabling fast real-time inference on mobile platforms without connectivity to the cloud. The mobile application is coded using Flutter, offering flexibility and robust performance on resource-limited smartphones with cross-platform support on Android and iOS. The basic model architecture is MobileNetV2, a convolution neural network that performs well in terms of both high accuracy and low computational needs, and this makes offline inference within feasible areas in remote fisheries environments.

To prepare datasets, annotation tools like LabelImg and CVAT are used to create accurate bounding boxes and multi-label annotations on fish images, which are a requirement for efficient model training. Data augmentation is performed using Keras' ImageDataGenerator, which simulates various visual transformations to enhance model resilience under various field and image conditions. Backend services via Node.js or Python Flask APIs can optionally be deployed for data storage and analytics across the network when remote network access is available. Development environment includes Python 3.x with Jupyter notebooks and TensorBoard for developing and training the model and visualizing the training, and Android Studio for developing and testing the mobile application for stability and performance on devices.

Figure 3.3: Technology Stack

3.3. Overall System Overview

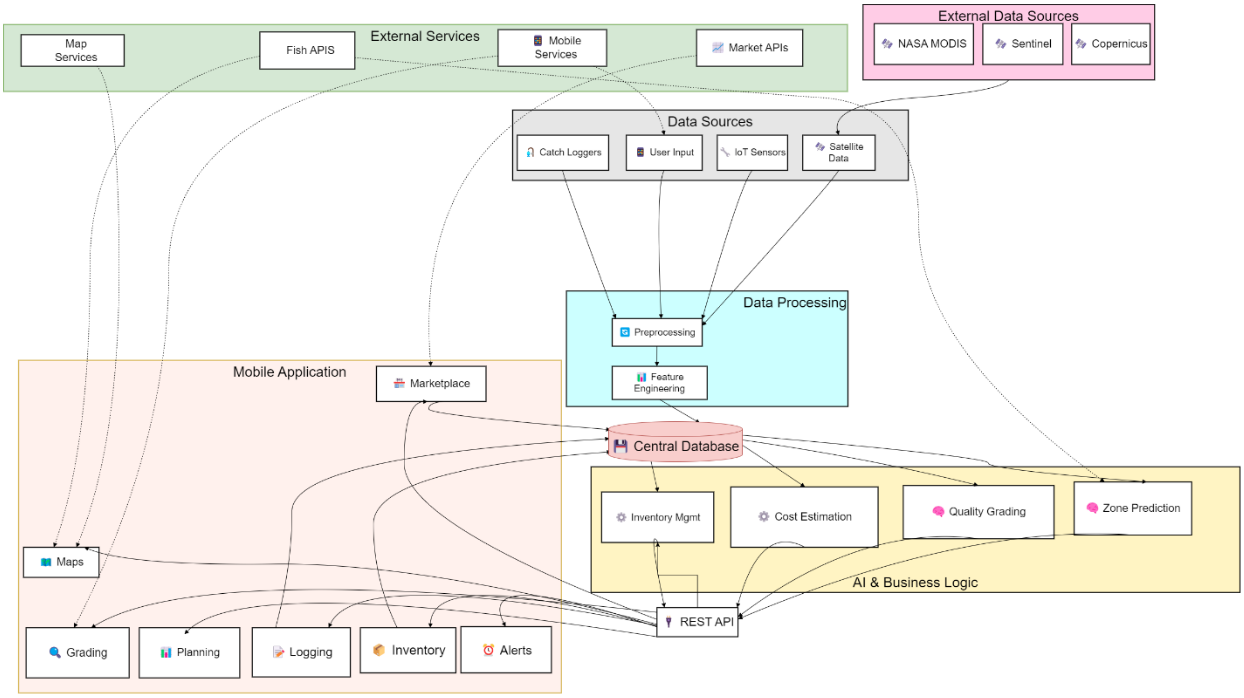

Figure 3.4: Overall System Overview Diagram

Our proposed system is based on an integrated, holistic architecture to address the multidimensional problems of Sri Lankan fishermen. The infrastructure is designed in distinct layers that facilitate seamless data flow from collection to decision-making information. Externally sensed satellite data from NASA MODIS, Sentinel provides underlying oceanographic parameters (SST and chlorophyll concentrations), which feed into our processing data layer alongside inputs from IoT sensors, catch loggers, and manual user input. This data undergoes rigorous preprocessing and feature engineering before being deposited in a centralized database. Business logic and AI layer is the brain of our system, comprising bespoke machine learning models for fish zone prediction (Random Forest/SVM/Neural Networks-based) and fish quality grading (Mobile Net CNN architecture with transfer learning-based), as well as business rules for dynamic trip cost calculation and inventory management. These components communicate through a RESTful API gateway that exposes processed intelligence to a mobile application, enabling features like interactive map-based zone visualization, AI-powered quality grading, trip planning functionality, digital catch logging, marketplace integration, inventory management, and license expiration alerts. The architecture engages in two-way communication with external services like mapping APIs (Google Maps/MapBox), market data feeds, mobile device services, and specialty fisheries APIs, creating an integrated decision-support ecosystem that transforms raw environmental and operational data into actionable intelligence for fishermen at both individual and commercial scales

3.4. Component overview

The fish quality grading system is modular in architecture to enable every functional unit to integrate in a seamless manner, hence providing an integrated solution perfectly suited for application on mobile devices. The process starts from data collection and annotation stage, where images of fish over calibration objects are gathered from landing sites and fish markets. The images are thoroughly annotated for size, physical damage, freshness, and species, thereby creating a detailed dataset crucial in enabling proper training of models.

Following data collection, preprocessing and data augmentation are conducted on the images. Normalization of image size, luminance, and contrast is done, as well as segmentation of the fish from backgrounds for visual representation. Rotation, flipping, and color augmentation are done to boost the dataset diversity and give the model the ability to be resilient to real-world variation in surroundings.

The prepared images are then analyzed by the feature extraction module to detect major anatomical features like the snout, tail, and body's widest points. Pixel measurements are calibrated to real world units using the calibration object as reference to enable precise morphometric data extraction necessary for accurate weight estimation and grading.

The core of the system is a number of AI models that are trained to do specific tasks: species and freshness classification, damage detection, and estimation of weight through regression analysis. The models make use of the annotated data and extracted features to make accurate appraisals for each fish specimen.

To attain best performance on handheld, the trained models quantized and pruned are transformed to TensorFlow Lite. This provides real-time efficient inference that can work offline and can be used to function within fisheries environments with poor connectivity.

The mobile application is the user interface, guiding the users through the image acquisition process, demonstrating object calibration inclusion, and showing the grading results in an easy-to-use multi-language user interface. It also supports optional features like data export and cloud synchronization.

Finally, the system also comprises a validation and feedback loop where AI predictions are continually compared with expert human graders and physical measurements. This process relies on user feedback and environmental metadata to continually refine and enhance the accuracy and reliability of the system, rendering the solution dynamic and responsive to field conditions.

Hierarchical, component-based design leads to the system being scalable, maintainable, and flexible, having been specifically made to address the challenges of Sri Lankan fisheries while prioritizing practical useability and scientific excellence.

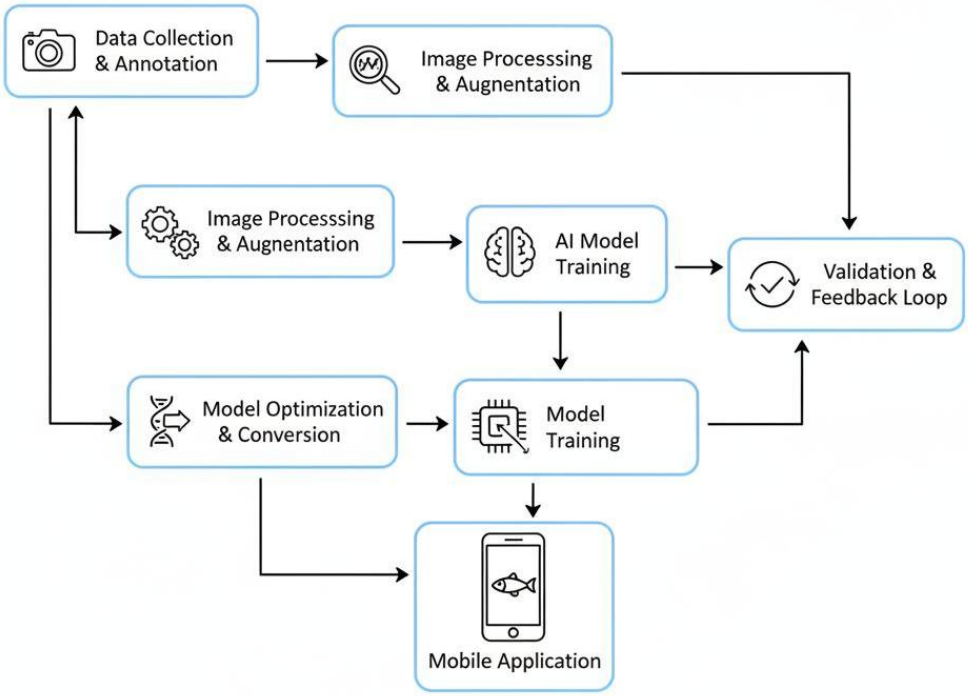

Figure 3.5: Component diagram

4. GANTT CHART

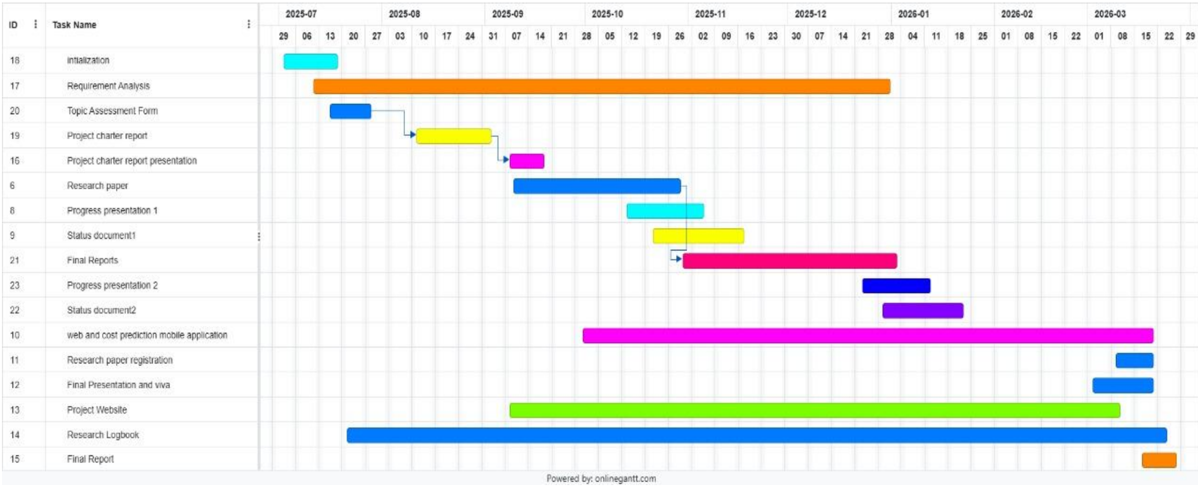

Figure 4.1: Gantt chart

5. WORK BREAKDOWN STRUCTURE

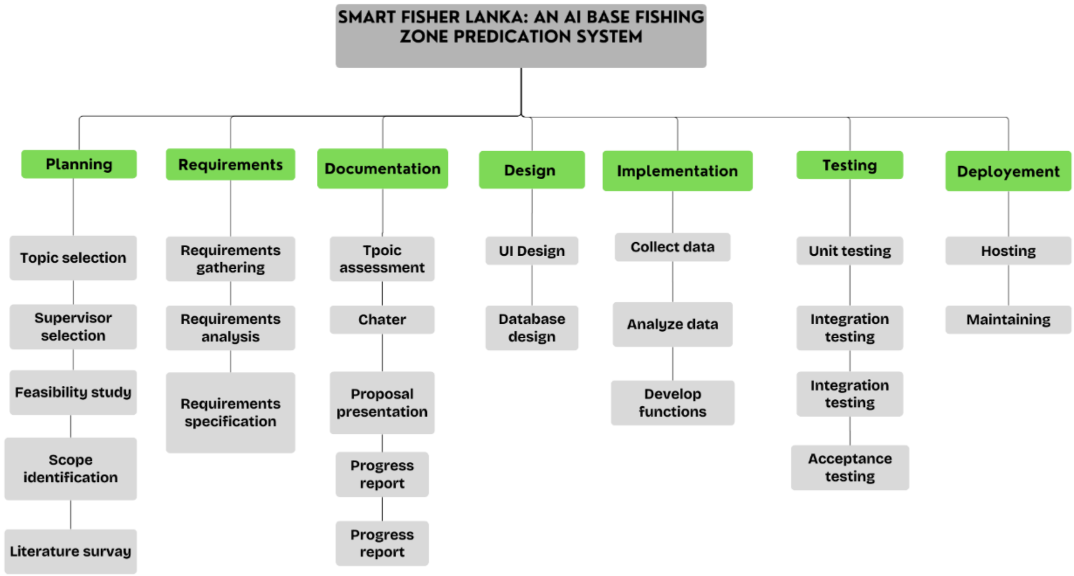

Figure 5.1: Work Breakdown Structure

6. DESCRIPTION OF PERSONAL AND FACILITIES

Facilitators:

- Prof. Pradeep Abeygunawardhena– Sri Lanka Institute of Information Technology
- Mr. Ashvinda Iddamalgoda– Sri Lanka Institute of Information Technology

7. Budget

Table 1:Budget

Task	Cost
Cloud Server (model training, backend & APIs)	$60 – $200 / year
Domain Name Registration	$10 – $15 / year
API Services (satellite data, maps, vision, notifications, etc.)	$25 (one-time)
Mobile App Publishing (Google Play Developer Account)	$50 – $150
Data Acquisition & Dataset Preparation (satellite, fish images, market data)	$50 – $150
IoT Devices & Field Equipment (GPS/catch loggers, testing tools)	$100 – $200
Field Testing & Validation (with fishermen, market trials)	$100 – $200
Contingency & Miscellaneous	$15 – $30
Total (First Year)	$400 – $1000

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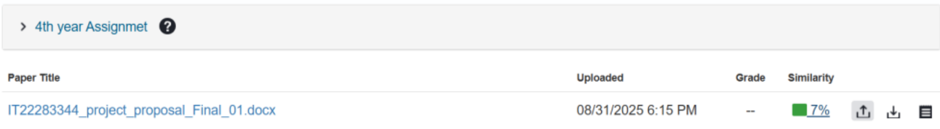
APPENDICES



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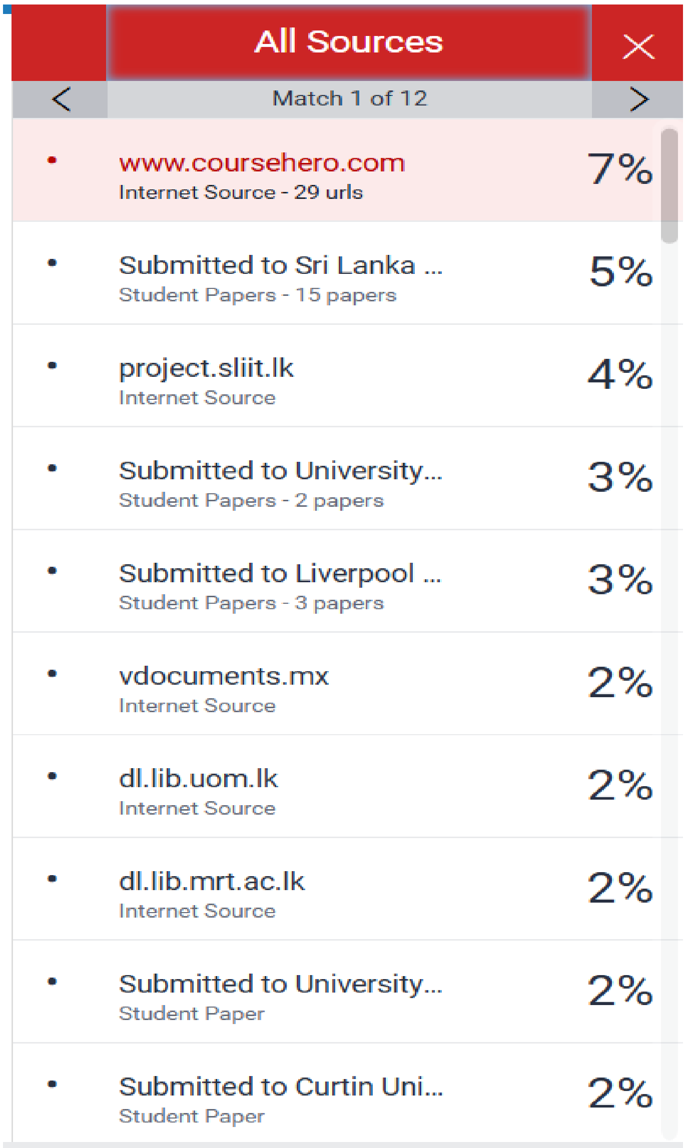

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